

Scientific Reviewer

12+3 · Accept with notes

Strengths: clean taxonomy, emb $\gg$ pixels + multi-block ablations, semantic+local probes, GPU-hour plots. Weaknesses: crop-as-aug, mixed 1%, qualitative RCDM, IN-centric.

Spine: Predict embeddings, not pixels · multi-block + EMA + narrow predictor · JEA|MAE|I-JEPA ·
→V-JEPA→2-AC (cite 2-AC for robots) · 54.2 / 66.9 $\gg$ 40.7 / $\sim 10\times$ / 72.4

Claim under review

What the paper actually asserts

- C1** Predicting in representation space with multi-block masking yields semantic ImageNet linear probes without hand-crafted view augmentations.
- C2** Multi-block $\gg$ random/block/raster (T6: 54.2 vs ≤ 20.2); emb targets $\gg$ pixels (T7: 66.9 vs 40.7).
- C3** Competes on local tasks (Clevr Dist 72.4) while closing the gap to view-invariance methods on semantics.
- C4** Scales efficiently: ViT-H/14 < 1200 GPU-h / $16 \times A100 < 72h$; $\sim 10\times$ vs MAE H/14.

Strengths

What earns Accept

Clean taxonomy

Fig. 2 JEA / Generative / JEPA + crisp Fig. 3 instantiation.

Isolating ablations

T6 multi-block; T7 emb $\gg$ pixels; T11 output-mask; T12 predictor depth.

Semantic + local

Reports both — rare honesty vs invariance-biased SSL.

GPU-hour plots

Figs. 1 & 5 — efficiency backed by compute, not only epochs.

No view-aug claim

Falsifiable recipe; open code aids reproduction.

Predictive framing

LeCun AMI / EBM grounding makes the 'why' legible.

Ablation evidence — thesis isolators

Authoritative numbers from [ablations.md](#)

T6 mask

multi-block 54.2 vs raster 15.5 / block 20.2 / random 17.6

ViT-B/16 300ep 1% lin

T7 space

emb 66.9 vs pixels 40.7

ViT-L/16; note epoch mismatch 500 vs 800

T8-10

target scale sweet spot; context ↑; #targets 1→4: 9.0→54.2

App. C

T11

output-mask 67.3 vs input-mask 56.1

ViT-H/16 300ep

T12

predictor depth 12→66.9 vs 6→64.0

ViT-L/16 500ep

Efficiency evidence — Fig. 5

1% IN vs GPU hours — persuasive but not iso-param

ViT-L/16 is trained for the equivalent of 600 IN1K epochs.

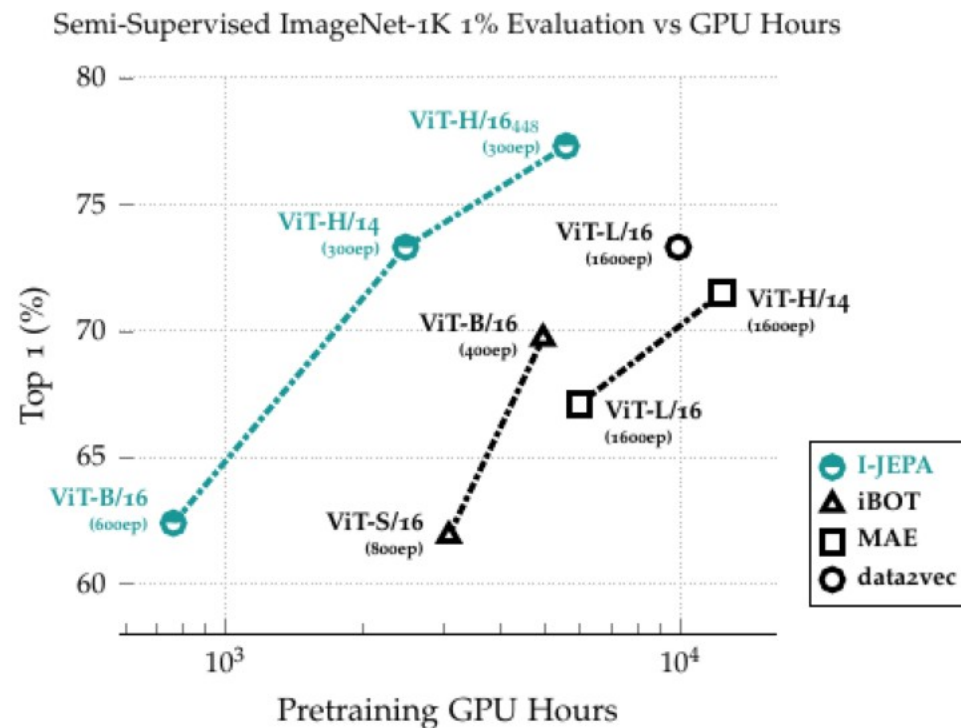


Figure 5. **Scaling.** Semi-supervised evaluation on ImageNet-1K 1% as a function of pretraining GPU hours. I-JEPA requires less compute than previous methods to achieve strong performance. Compared to MAE and data2vec, I-JEPA obtains a significant

Reviewer note: mix of arch/epochs across methods — directionally strong, demand matched-GPU-hour curves for reuse.

Weaknesses — ask-for-revision style

Punish soft spots without rejecting the thesis

Crop-as-aug philosophy

'No hand-crafted augs' still uses random resized crop-like block sampling as the sole inductive bias — soft augmentation. Clarify the line.

Mixed 1% protocol

Table 2 uses FT or linear 'whichever works best' — hurts strict ranking.

Qualitative RCDM only

Fig. 6 suggestive; no quantitative predictor calibration / uncertainty probe.

IN-centric ablations

T6-T12 almost all IN-1% linear — 'semantic' $\approx$ ImageNet class features; robot OOD untested.

data2vec match

Closest concurrent comparison could be tighter on matched compute.

Collapse theory thin

EMA 'essential' empirically; limited analysis without it.

Crop-as-aug — philosophical pressure point

Reviewer question to put on the authors

Is multi-block sampling a masking inductive bias (authors) or a soft spatial augmentation that smuggles crop-style invariance (reviewer)? Paper argues targets are prediction targets not views — demand a sharper criterion: if you cannot train without block sampling, how is that different from requiring RandomResizedCrop?

Internal reuse answer

Treat mask geometry as a hyperparameter with a documented sweet spot (T8-10), not as 'augmentation-free' marketing. Empiricist freezes defaults before other axes.

Internal reuse checklist

Before citing I-JEPA in our work

☐ Reproduce Table 7 (pixel vs embedding loss) on our hardware — direction, not absolute %.

☐ Matched-GPU-hour curve vs MAE for our preferred backbone.

☐ Stress-test multi-block scales outside (0.15, 0.2) — hypersensitivity is real (T8).

☐ When citing for robotics: cite V-JEPA 2-AC, not this paper's experiments.

☐ Footnote eval protocol: no [CLS]; avg-pool target-encoder — differs from DINO/iBOT.

☐ Pin 1% ImageNet split / seed if claiming low-shot numbers.

Robotics citation rule

Do not launder 2-AC results into this PDF

CLEAR

This paper: predictor-as-world-model stub (positional mask tokens); LeCun AMI framing; no robot experiments.

Robot SR / Franka / DROID / CEM / image-goal planning → V-JEPA 2-AC, arXiv 2506.09985.

Reviewer red flag

Any talk slide that puts a Franka success rate next to 'I-JEPA (Assran 2023)' without 2-AC is a citation error.

Verdict

Accept landmark method paper; punish overclaim

ACCEPT

Clean taxonomy + isolating ablations + dual probes + compute plots. Thesis (emb prediction + multi-block) is earned.

WITH NOTES

Clarify crop-as-aug; tighten 1% protocol; add quantitative predictor probes; stop robotics laundering — cite 2-AC.

Exam bite: Accept the method; punish 'augmentation-free' marketing and any robot-SR cite to this PDF alone.

Numbers tattoo — pin these

Shared across all five seats

79.3%

ViT-H/14 linear

73.3%

H/14 · 1% labels

81.1% / 77.3%

H/16@448 lin / 1%

54.2

multi-block (T6)

66.9 >> 40.7

emb >> pixels (T7)

72.4

Clevr Dist (T4)

~10×

cheaper than MAE H/14

<72 h

16×A100 H/14

<1200

GPU-hours H/14

87.1 vs 87.8

FT H/16@448 vs MAE

no view augs

single-view JEPA

2-AC not here

robots → cite 2506.09985

Taxonomy figure — reviewer's friend

Fig. 2 makes the claim diagram-checkable

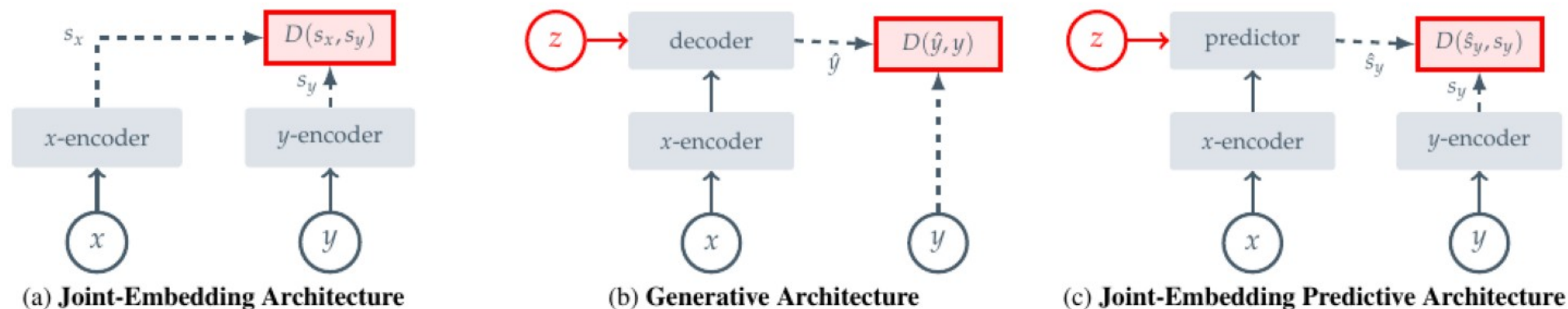


Figure 2. Common architectures for self-supervised learning, in which the system learns to capture the relationships between its inputs. The objective is to assign a high energy (large scalar value) to incompatible inputs, and to assign a low energy (low scalar value) to compatible inputs. **(a)** Joint-Embedding Architectures learn to output similar embeddings for compatible inputs x, y and dissimilar embeddings for incompatible inputs. **(b)** Generative Architectures learn to directly reconstruct a signal y from a compatible signal x , using a decoder network that is conditioned on additional (possibly latent) variables z to facilitate reconstruction. **(c)** Joint-Embedding Predictive Architectures learn to predict the embeddings of a signal y from a compatible signal x , using a predictor network that is conditioned on additional (possibly latent) variables z to facilitate prediction.

resulting representations are typically of a lower semantic level and underperform invariance-based pretraining in off-the-shelf evaluations (e.g., linear-probing) and in transfer

approaches on semantic tasks and achieves better performance on low-level vision tasks such as object counting and depth prediction (Sections 5 and 6). By

Demand authors keep this three-way distinction — do not blur JEPA into 'just another MAE'.

Scientific Reviewer — close

Takeaways + exam bite

1. Strengths: taxonomy, T6/T7 isolators, semantic+local, GPU-hour plots.
2. Weaknesses: crop-as-aug, mixed 1%, qualitative RCDM, IN-centric.
3. Reuse checklist before citing; matched compute vs MAE.
4. Robotics → cite V-JEPA 2-AC, not this paper.

EXAM BITE

Accept landmark JEPA image paper; punish aug-free overclaim and robotics citation laundering.